ROLLUPOS APPLICATION USER MANUAL • FIELD & PRODUCTIVITY

🧭 CSC Site Discovery

Offline-First HVAC/BAS Field Survey Companion • Division 23 Checklists, CLAHE Nameplate OCR & Building Mapper

Version: v3.0.0 | Build Hash: #3e6a4ddf | 🔒 Per-User Authorized Cartridge | Auto-Synced: 2026-10-26 16:54

🆕 **What Changed in This Build:** Auto-detected new interactive controls in v3.0.0: 📋 Units (6), 🗺️ Map, ☀️, ⚙️, Recalibrate GPS, All Units, 🔴 Caution / Replace, 🟢 Good, 📋 Clone Last +1, 📷 Snap Wall Map, 🖼️ Load Plan, 🗑️, ➕ Area/Floor, 📍 Move Pin, 📸 + SNAP NEW UNIT

1. ⚡ 60-Second Quickstart Guide

1 **Initialize Your Site Walk (`⚙️ Walk / API Vault`)**
Tap `⚙️ Walk / API Vault` (or `Walk / Export`) to enter the Facility Name, CSC Job/Estimate Number (`CSC-26-xxxx`), and optionally save a personal Gemini API key in your private on-device Vault.

2 **Add Equipment Stations & Select System Type**
Create stations for AHUs, RTUs, Chillers, Boilers, Hydronic Pumps, VAVs, or Main BAS TCPs. The app dynamically loads the matchingCSC Division 23 (`RFI-23-001` through `RFI-23-011`) engineering checklist.

3 **Capture 3-Slot Burst Photos w/ Stamped-Metal CLAHE**
Snap (1) Wide Equipment Context, (2) Rusted/Stamped Nameplate (auto-enhanced via local CLAHE contrast boosting + serial vintage decoding), and (3) Inside Control Cabinet / Termination Strip.

4 **Pin Evacuation Floorplan & Export Survey Package**
Snap the wall-mounted Building Evacuation Map, tap to drop equipment pins with GNSS/Compass bearings, record 150Hz high-pass filtered voice walkthroughs, and export the complete JSON/CSV package via Android Share Sheet.

2. 🧭 Screen-by-Screen & Navigation Map

SCREEN / TAB / MODAL	WHAT IT CONTROLS & HOW TO USE IT
📋 **Equipment Station List (Left/Top)**	Add, filter, and switch between rooftop, mechanical room, and terminal unit stations during a walk.
🔍 **Multimodal Station Inspector**	Dynamic RFI-23-001..011 checklists, 3-slot CLAHE camera capture, serial age decoder, and 150Hz filtered voice notes.
🗺️ **4-Layer Building & Evac Mapper**	Pin equipment stations directly onto photographed floorplans/evacuation maps with compass orientation and Laplacian video sharpness scoring.
🧮 **Field Valve / VA / Labor Sizers**	On-site deterministic Cv valve sizer (4–5 psi ΔP / β ≥ 0.50), NEC Art. 725 80% Class 2 panel VA sizer, and rough-in labor estimator.
⚙️ **Session Manager, Vault & Export**	Manage walk metadata, optional Gemini OCR key, Golden Demo Walk loader, and 1-tap JSON/CSV exports.

3. 🔬 In-Depth Feature Guide & Workflows

CSC Overlay & Safety Checklist Enforcement (`RFI-23-001`..`011`)

Prompts field engineers to verify factory OEM BACnet cards (`Overlay, Never Gut`), hardwired VFD HOA/speed split (`RFI-23-006`), DPDT manual-reset serpentine freeze stats (`RFI-23-007`), Div 23 21 13 inline valve/well boundaries (`RFI-23-005`), and dedicated Div 26 120VAC 15A TCP circuits (`RFI-23-009`).

CLAHE Stamped-Aluminum Nameplate Enhancer & Offline Queue

Applies local canvas Contrast-Limited Adaptive Histogram Equalization so sun-bleached or rusted outdoor RTU/chiller data plates become legible instantly—working 100% offline in mechanical basements.

150Hz Mechanical-Room High-Pass Audio Filter

Uses WebAudio high-pass filtering to strip low-frequency fan/pump rumble (`<150Hz`) from walk-and-talk voice memos for crystal-clear Whisper transcription back at the office.

4. ⚙️ Settings, Secret Vault & Multi-User Data Isolation

SETTING KEY	FIELD DESCRIPTION	TYPE	DEFAULT
`default_job_number`	Default CSC Job / Estimate Number	`text`	`CSC-26-NEW`
`engineer_name`	Lead Engineer / Estimator Name	`text`	`CSC Field Engineer`

🔒 **Storage Namespace Pattern:** `rollup::v1::u::<google-email>::app::csc-site-discovery::*`

📦 **Active App Storage Keys:** `__user_config__` (Managed automatically)

🔐 **Private On-Device Vault Secrets:** None required (uses Hub session)

Isolated Per-User Storage (RollupBridge) · Mobile Camera / Multimodal Capture · WebAudio / Voice Recording Pipeline · Live GNSS / Compass Telemetry · Android Share Sheet & CSV/JSON Export · Encrypted On-Device Secret Vault

5. 🎛️ Live UI Controls Reference & Recent Updates

Interactive Controls Detected in Current Build (`v3.0.0`):

📋 Units (6) | 🗺️ Map | 🌞 | ⚙️ | Recalibrate GPS | All Units | 🔴 Caution / Replace | 🟢 Good | 📋 Clone Last +1 | 📷 Snap Wall Map | 🖼️ Load Plan | 🗑️ | ➕ Area/Floor | ➕

Pro Tips & Best Practices:

- Always enforce NFPA 70E Lockout/Tagout (LOTO) and PPE before opening live control cabinets for internal termination photos.
- No cell signal in a sub-basement chiller plant? Leave the API key blank—every photo, checklist, and voice clip saves to IndexedDB/RollupBridge offline.

Recent Cartridge Update Log (Auto-Tracked on Publish):

- `fe98244` (2026-10-26) — RollupOS v1.1.0 + CSC Site Discovery v2.0.0 (2026-10-26 15:13)
- `9ac185f` (2026-10-24) — RollupOS Hub & Mini-App sync (2026-10-24 22:19)
- `22d612c` (2026-10-24) — RollupOS Hub & Mini-App sync (2026-10-24 19:59)
- `0c21c6e` (2026-10-24) — RollupOS Hub & Mini-App sync (2026-10-24 16:38)

RollupOS Automated Documentation Engine • Cartridge ID: `csc-site-discovery`

Canonical PDF: `manuals/csc-site-discovery_User_Manual.pdf`